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Air Quality Forecasting and Alert System

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Agenda

- Introduction
- Motivation (Why this title)
- Problem Definition
- Objectives
- Data set Description
- Risk Analysis
- Tools and Techniques used
- Phase I : Machine Learning Model and its analysis
- Phase II : Building MLOPs



1. Introduction

Air pollution is a serious environmental issue that impacts human health, climate, and ecosystems. Rapid industrialization, urbanization, increased vehicle emissions, and population growth have led to poor air quality in many cities. This deterioration in air quality causes major health problems such as respiratory diseases, cardiovascular issues, and premature deaths, making air pollution monitoring and control extremely important.

With advancements in Machine Learning (ML), historical air quality data can be analyzed to predict future pollution levels and provide early warnings. However, to ensure accuracy, reliability, and scalability, MLOps practices are essential. This project integrates machine learning models with MLOps tools like ZenML to develop an Air Quality Forecasting and Alert System that continuously monitors data, improves predictions, and generates timely alerts when air quality exceeds safe limits, supporting better decision-making and public health protection.



2. Motivation (Why this Title?)

- Rising Air Pollution: Urban air quality is deteriorating rapidly.
- Health Protection: Early air quality alerts help people take preventive measures.
- Limits of Traditional Systems: Existing systems are reactive, not predictive or automated.
- Early alerts can help citizens reduce exposure to polluted air
- There is a need for a proactive system rather than a reactive one



3. Problem Definition

- Existing air quality systems mainly report current pollution levels and lack accurate future forecasting.
- Alert mechanisms are mostly manual or reactive, reducing their real-world usefulness.

Problem Statement:

To design and implement a machine learning–based air quality forecasting and alert system using an end-to-end MLOps pipeline that ensures reliability, scalability, and continuous deployment.

4. Objectives

- To ingest and preprocess historical air quality data for reliable modeling.
- To develop, train, and compare multiple machine learning models and select the best-performing one.
- To forecast future AQI levels rather than only reporting current air quality.
- To automatically generate alerts when AQI exceeds unsafe thresholds.
- To build an end-to-end, scalable MLOps pipeline with model versioning and monitoring using ZenML



5. Dataset Description (Kaggle)

link: [Air Quality Data in India \(2015 - 2020\)](#).

Dataset Source: Kaggle – Air Quality Dataset

Category	Details
Dataset Type	Structured tabular dataset
Total Records	707,876 Records
Total Features	6 attributes
Input Features	PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2
Target Variable	AQI
ML Task Type	Regression



6. Risk Analysis

Risk	Description	Mitigation Strategy
Data Quality	Missing or inconsistent data	Data cleaning & imputation
Model Bias	Poor generalization	Model comparison & validation
Overfitting	Complex models overfit training data	Cross-validation
Data Drift	Changing pollution patterns	Periodic retraining via MLOps
Alert Errors	False positives/negatives	Threshold tuning & evaluation

7. Tools and Techniques Used

MLOps Tool

- 1.ZenML
- 2.DVC
- 3.ML flow
- 4.Docker

Programming Language

- Python

Libraries

Pandas – Data handling
NumPy – Numerical computation
Scikit-learn – ML algorithms
XGBoost – Gradient boosting
Matplotlib / Seaborn – Visualization

Phase I – ML Life Cycle

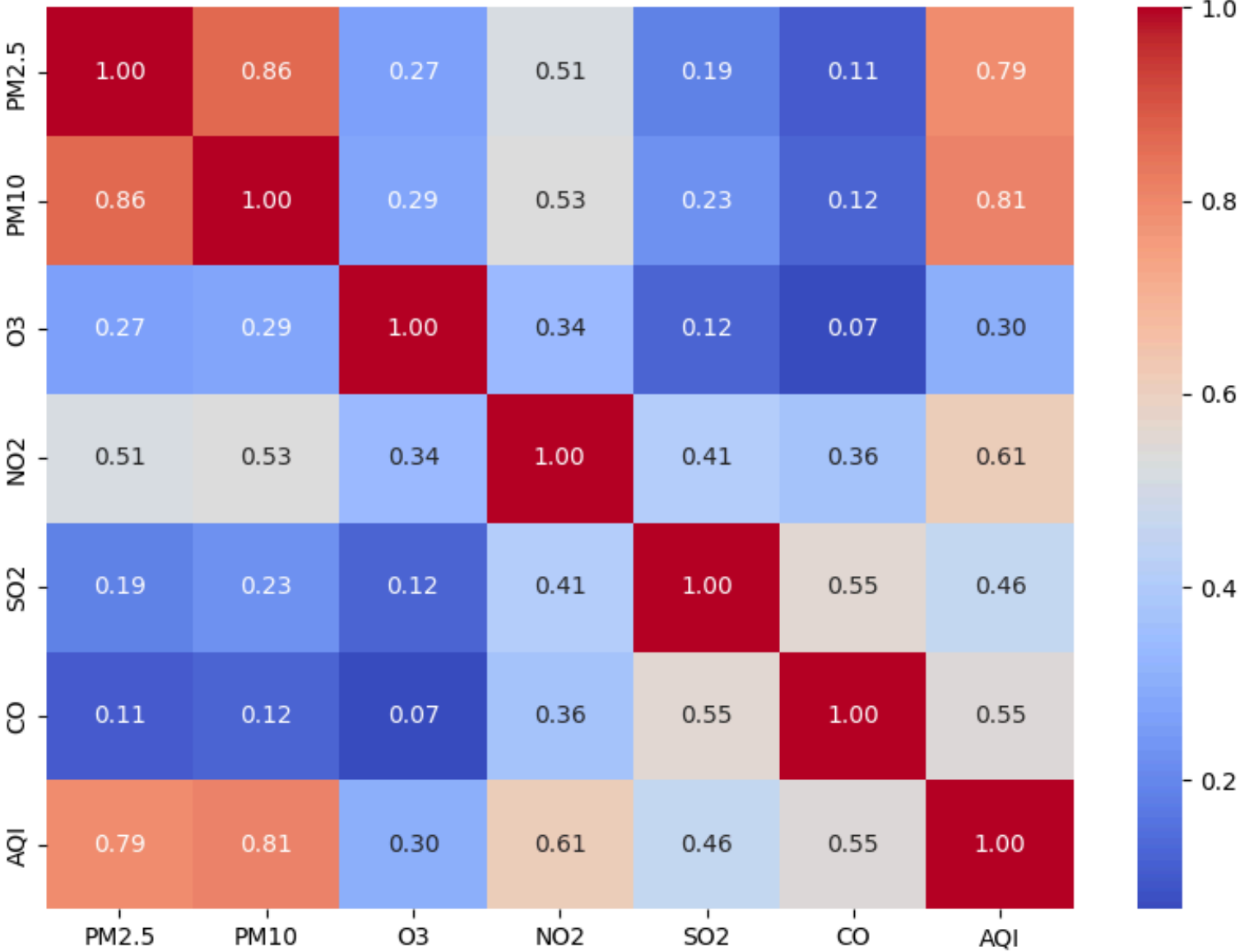
1.Exploratory Data Analysis

- Exploratory Data Analysis was performed to understand the air quality dataset and its characteristics.
- The dataset contains air pollutant parameters such as PM2.5, PM10, CO, O₃ and AQI.
- Missing values were identified in multiple columns.
- Data type validation was performed to ensure all pollutant values were numerical.
- Basic statistical analysis (mean, min, max) was used to understand pollutant concentration ranges.
- Correlation analysis was carried out to observe the relationship between pollutant levels and AQI.
- Outliers and invalid entries were handled during the data cleaning phase using imputation techniques.

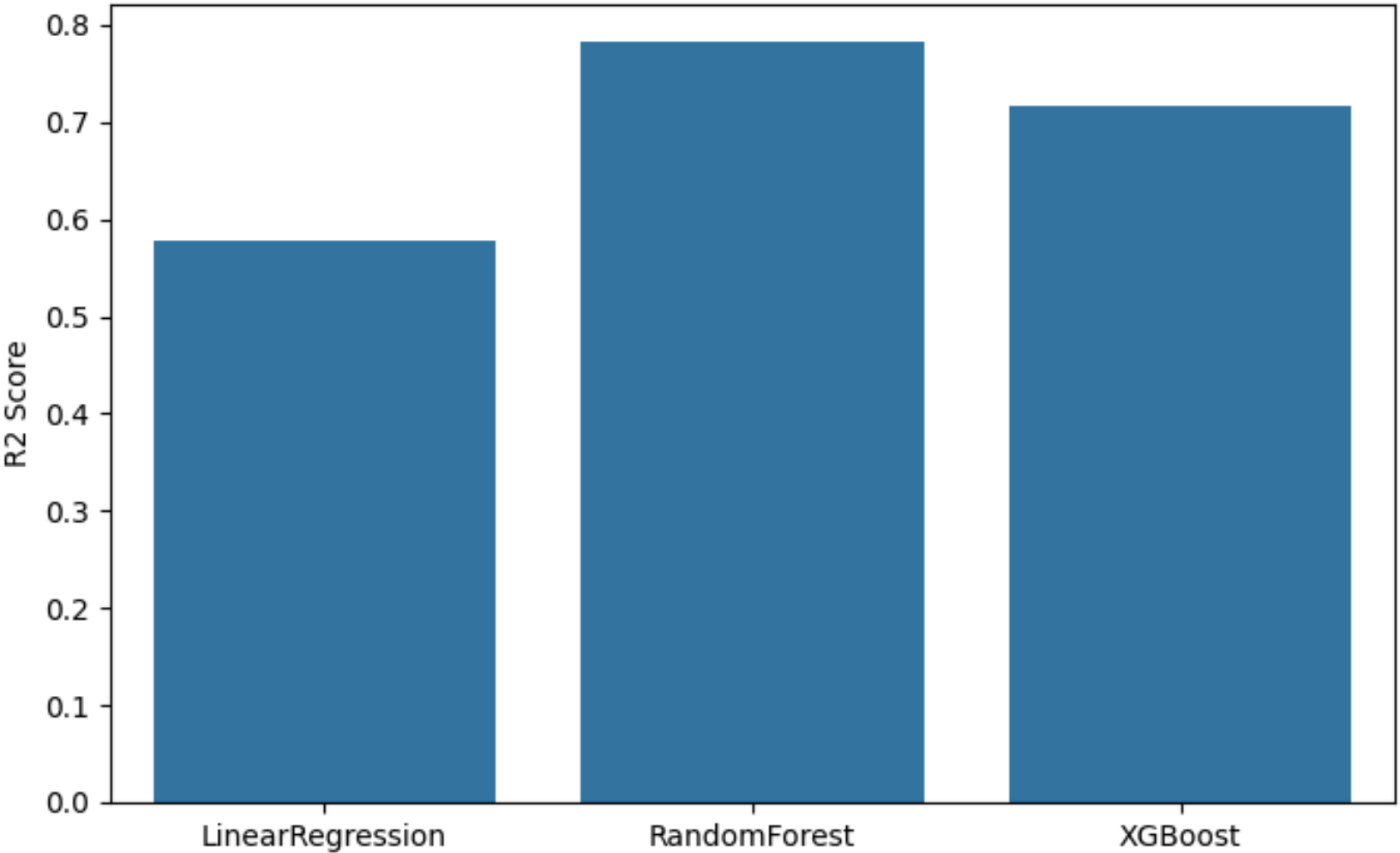


1.Exploratory Data Analysis

Feature Correlation with AQI



Model R2 Scores Comparison





2. List of algorithms chosen to build ML model

1. Linear Regression

- Used as a baseline model
- Simple and interpretable

2. Random Forest Regressor

- Ensemble-based model
- Handles non-linearity and feature interactions well

3. XGBoost Regressor

- Gradient boosting algorithm
- Provides high performance and feature importance scores

3. Metrics used to evaluate the ML model

1. Root Mean Squared Error (RMSE)

- Measures the average magnitude of prediction error.

2. R² Score (Coefficient of Determination)

- Indicates how well the model explains variance in AQI values.



4. Evaluation of ML Model

- Each model (Linear Regression, Random Forest, XGBoost) was trained on the cleaned dataset.
- Predictions were compared against actual AQI values.
- RMSE and R^2 scores were calculated for each model.
- XGBoost achieved lower RMSE and higher R^2 score compared to other models.

5. Identification of Best Model

Based on evaluation results:

Random Forest was identified as the best model

Reasons for selection:

- Lowest prediction error (RMSE)
- Highest R^2 score
- Better handling of non-linear relationships
- Built-in feature importance for feature selection
- Suitable for deployment in MLOps pipelines



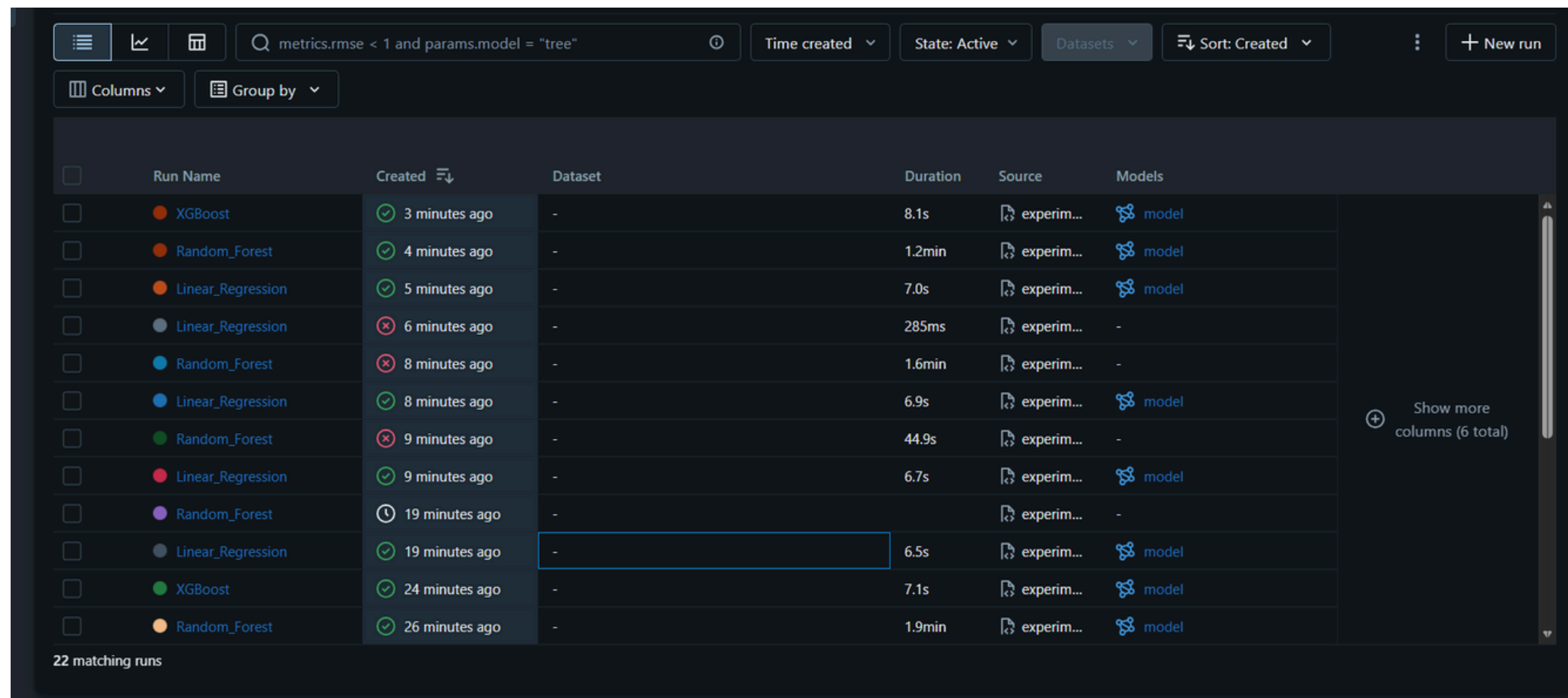
	Component	MLOPs Tools
1	Data Management	DVC
2	Experiment Tracking and Version Control	MLflow
3	Automation and Pipeline Orchestration	ZenML
4	Model Deployment & Serving	Docker(streamlit)
5	Monitoring & Governance and collaboration	Evidently AI

MLOPs Tools: ML Flow

- Tracks the historical air quality dataset (city_hour.csv) used for model training.
- Maintains history of changes in the dataset to ensure reproducibility across experiments.
- Ensures that each model (Linear Regression, Random Forest, XGBoost) uses the same dataset version for training and evaluation.
- Prevents data leakage between training and test sets.
- Supports reproducible ML experiments, enabling fair comparison of different models and configurations.
- Logs model hyperparameters, e.g.:
 - n_estimators, max_depth for Random Forest
 - learning_rate, n_estimators for XGBoost
- Logs metrics such as R^2 and MSE for all models.

MLOPs Tools: ML Flow

- Facilitates easy comparison of model performance:
- Linear Regression (baseline model)
- Random Forest (bagged tree ensemble)
- XGBoost (boosted tree ensemble)
- Helps select the best model based on highest R^2 and lowest MSE for AQI prediction.
- Ensures experiment reproducibility and traceability across multiple runs, supporting reliable air quality forecasting and alert generation



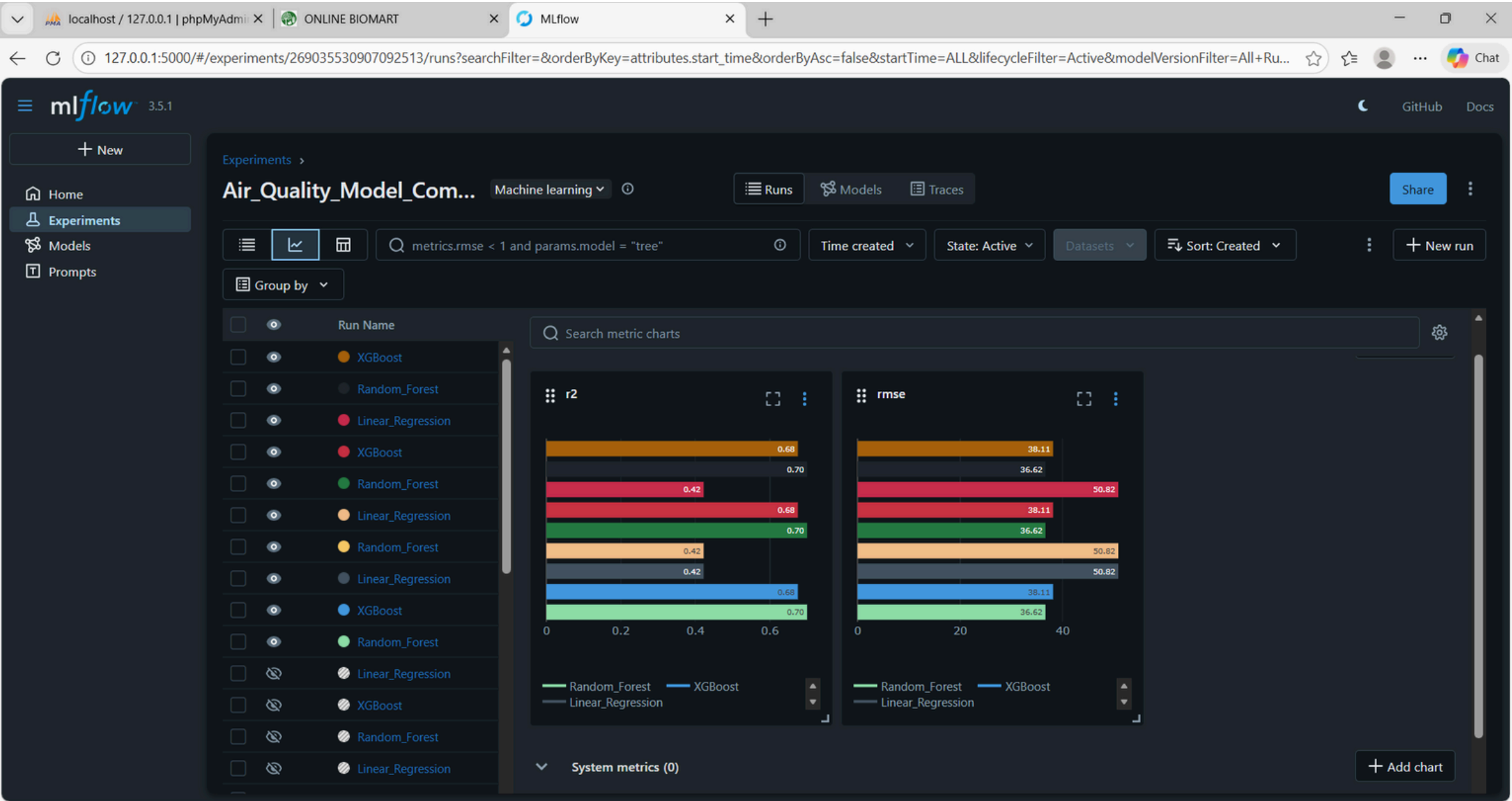
☐	Run Name	Created ↕	Dataset	Duration	Source	Models
☐	XGBoost	✓ 3 minutes ago	-	8.1s	experim...	model
☐	Random_Forest	✓ 4 minutes ago	-	1.2min	experim...	model
☐	Linear_Regression	✓ 5 minutes ago	-	7.0s	experim...	model
☐	Linear_Regression	✗ 6 minutes ago	-	285ms	experim...	-
☐	Random_Forest	✗ 8 minutes ago	-	1.6min	experim...	-
☐	Linear_Regression	✓ 8 minutes ago	-	6.9s	experim...	model
☐	Random_Forest	✗ 9 minutes ago	-	44.9s	experim...	-
☐	Linear_Regression	✓ 9 minutes ago	-	6.7s	experim...	model
☐	Random_Forest	⌚ 19 minutes ago	-		experim...	-
☐	Linear_Regression	✓ 19 minutes ago	-	6.5s	experim...	model
☐	XGBoost	✓ 24 minutes ago	-	7.1s	experim...	model
☐	Random_Forest	✓ 26 minutes ago	-	1.9min	experim...	model

22 matching runs

Show more columns (6 total)



MLOPs Tools: ML Flow





MLOPs Tools: DVC

- Tracks dataset versions for the Air Quality Forecasting system (city_hour.csv)
- Maintains history of changes in the training dataset
- Ensures each model (Linear Regression, Random Forest, XGBoost) uses the exact same dataset version as previous runs
- Prevents data leakage across experiments
- Enables reproducible ML experiments, especially when testing new models or feature improvements
- Supports collaboration, connecting dataset versions with MLflow-tracked model versions

MLOPs Tools: Zen ML

1. Ingest Data

- Loads the air quality dataset (final.csv)
- Splits data into training and testing sets
- Ensures consistent input data for model training

2. Clean Data

- Handles missing values
- Removes irrelevant or duplicate records
- Ensures data is suitable for machine learning models

3. Train Model

- Trains regression models on cleaned data
- Models used:
 - Linear Regression
 - Random Forest
 - XGBoost
- Selects the best-performing model based on evaluation metrics

4. Evaluate Model

- Evaluates trained models using:
 - Mean Squared Error (MSE)
 - R^2 Score
- Compares model performance on test data
- Helps identify the most accurate model

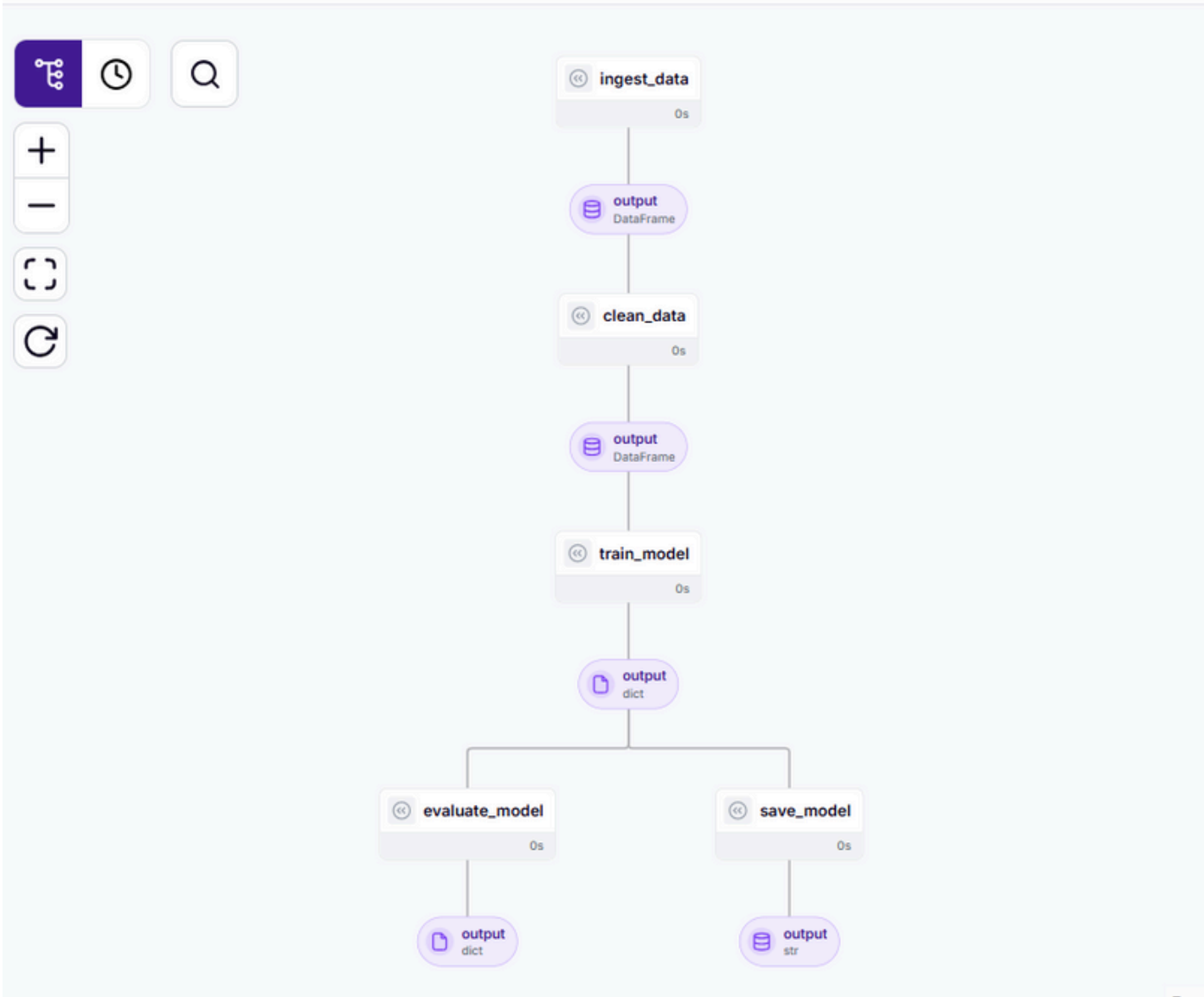
5. Save Model

- Saves the best-trained model
- Stores model for reuse in:
 - Forecasting
 - Real-time AQI prediction
 - Web application integration



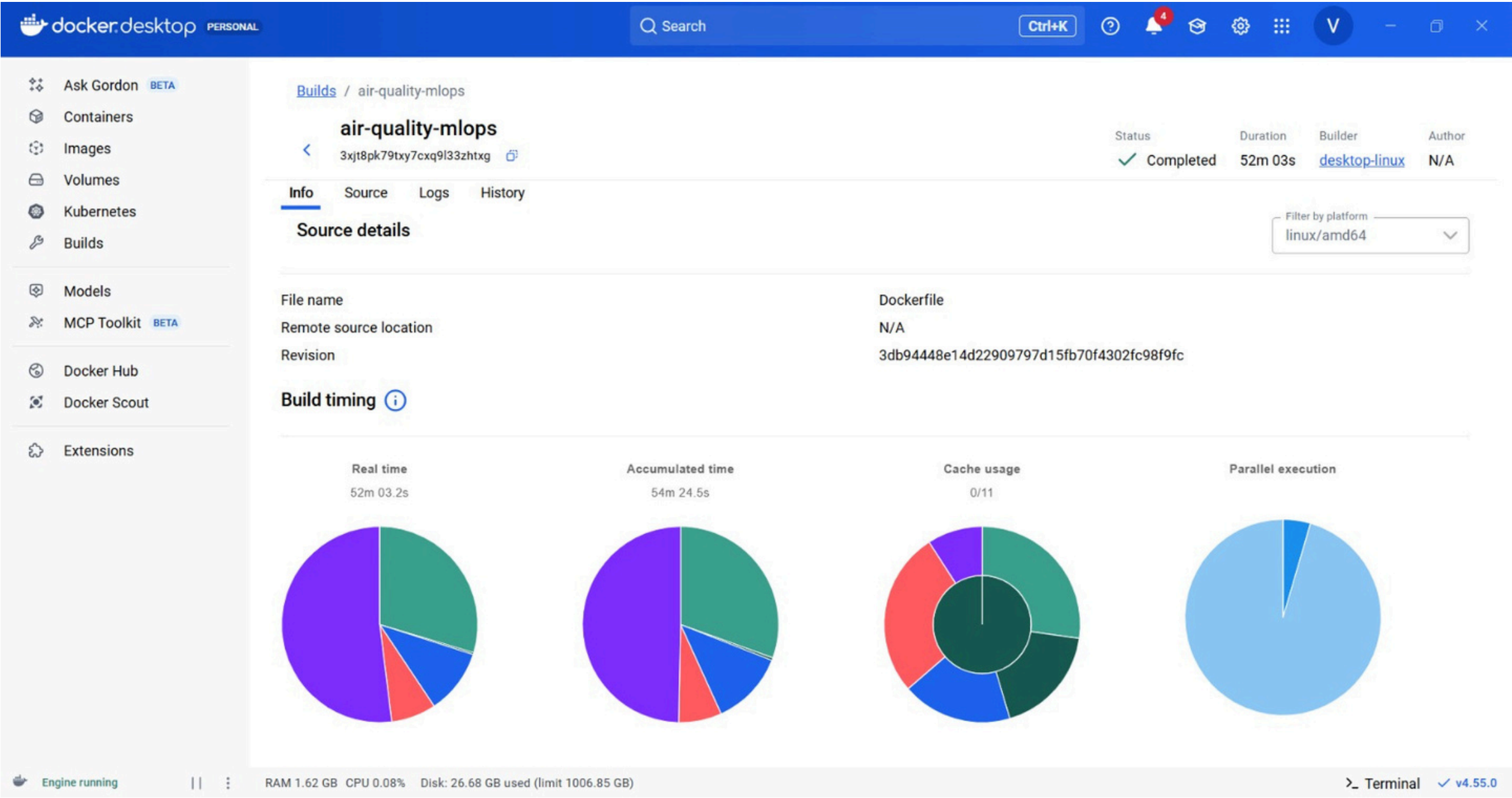
MLOPs Tools: Zen ML

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Zen ML pipeline

MLOPs Tools: Docker



Zen ML pipeline



MLOPs Tools: Docker

- Docker bundles our model, code, and dependencies into one container.
- This guarantees the project runs the same way on any machine, avoiding setup headaches.
- It solves issues like library mismatches or version conflicts.
- Docker also makes deployment quick, scalable, and reliable.
- In short, it ensures our model is easy to deploy and runs smoothly everywhere.



MLOPs Tools: Evidently AI

- Evidently AI is a tool to monitor ML models in production.
- It tracks data drift and performance changes to make sure predictions stay accurate.
- Whenever the model starts making errors, it indicates retraining is necessary.
- This helps maintain a stable and dependable system.
- EvidentlyAI monitors the air quality prediction model after deployment.
- Detects changes in input data, like pollution levels or weather conditions.
- Identifies data drift and drop in prediction accuracy.
- Alerts when the model's forecasts become unreliable.
- Indicates when retraining of the model is needed.
- Helps maintain accurate and dependable air quality predictions over time.

MLOPs Tools: Evidently AI

Drift is detected for 40.0% of columns (2 out of 5).

Search						
Column	Type	Reference Distribution	Current Distribution	Data Drift	Stat Test	Drift Score
> AQI	num			Detected	Wasserstein distance (normed)	0.147562
> PM2.5	num			Detected	Wasserstein distance (normed)	0.123252
> CO	num			Not Detected	Wasserstein distance (normed)	0.078003
> NO2	num			Not Detected	Wasserstein distance (normed)	0.074952
> PM10	num			Not Detected	Wasserstein distance (normed)	0.022228

Drift detection



Final Dashboard

Deploy

Air Quality Forecasting and Alert System

Current Air Quality Input

PM2.5

10.00

-

+

PM10

10.00

-

+

NO2

10.00

-

+

CO

10.00

-

+

Predicted AQI: 68.80

AQI Status

Moderate

Pollutant Levels Visualization

Department of AI and ML

<#>

Conclusion

The Air Quality Forecasting and Alert System provides accurate, real-time predictions of air pollution levels to ensure environmental safety and timely alerts. By integrating DVC for data versioning, MLflow for experiment tracking, and ZenML for orchestrating reproducible ML pipelines, the system ensures structured and maintainable workflows. Docker enables consistent deployment across different environments, while EvidentlyAI continuously monitors model performance, detects data drift, and triggers alerts for retraining when necessary. Overall, the project demonstrates a robust, reliable, and scalable approach to forecasting air quality, combining modern ML operations tools to maintain high accuracy and efficiency over time.

Drift detection